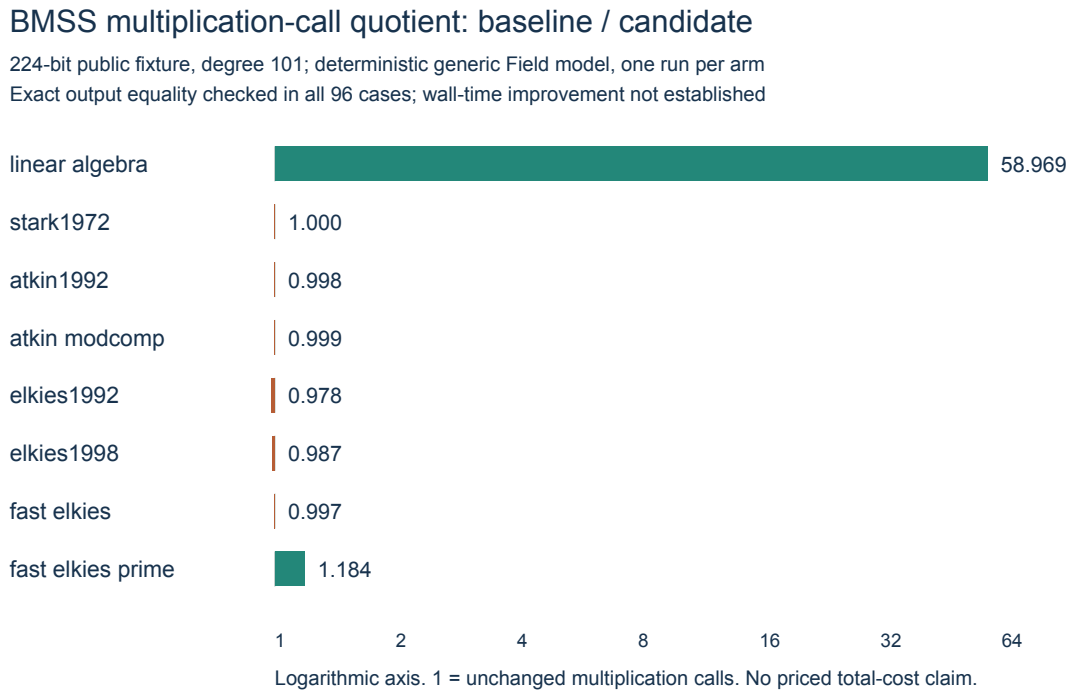
BMSS constructor optimization

2026-10-14 · Native exact-output evidence

All eight BMSS methods at 192, 224, 256 and 511 bits, degrees 7, 31 and 101: 96 exact baseline/candidate output matches. Kernel, codomain and rational map coefficients agree with the Kohel reference. Four public point transports are checked per constructor case.

Arithmetic counts at 224 bits, degree 101

Algorithm	Baseline mul	Candidate mul	Quotient	Baseline inv	Candidate inv
linear algebra	273659435	4640748	58.969	611	209
stark1972	10853630	10856762	1.000	1151	109
atkin1992	547470	548379	0.998	409	109
atkin modcomp	627414	628032	0.999	417	213
elkies1992	20109	20565	0.978	255	106
elkies1998	11127	11277	0.987	249	200
fast elkies	46958	47108	0.997	108	59
fast elkies prime	1073345	906695	1.184	110	110



The quotient counts generic Field multiplication calls. Inversions are opaque and counted separately. Setup, preparing sigma, independent reference construction and verification are excluded. Counts are deterministic for these fixed inputs; their model is not a CPU instruction count or a priced total cost.

Changes and limits

Row-independent series products are computed once per column. The second rung batches fixed series and Newton denominators, with direct division retained at small sizes. The third uses forward elimination followed by back substitution. Raw receipts distinguish the rungs and retain any additional multiplications incurred by batching.

This shared Apple M4 Pro host has 14 logical cores and 48 GiB RAM. Wall time is unqualified; no wall samples are promoted. Qualified multi-CPU A/A and A/B comparisons, the other inventory families, named NIST whole-search/replay totals, and actual release publication remain open. The 2,000-additional-map search remains separate and unfinished.

Comparison SHA256: 678d9a8974c3324fec62e74d1adc8011b63efb69379e3804ee19c6136ac5e906 Baseline and candidate run digests and complete case inputs are in COMPARISON.json and the source-linked raw JSONL records. Read PROTOCOL.md, INVENTORY.md and ATTEMPTS.md for boundaries and censored attempts.